

Questions 1 and 2 refer to the electron micrograph, which shows a cell from the pancreas.

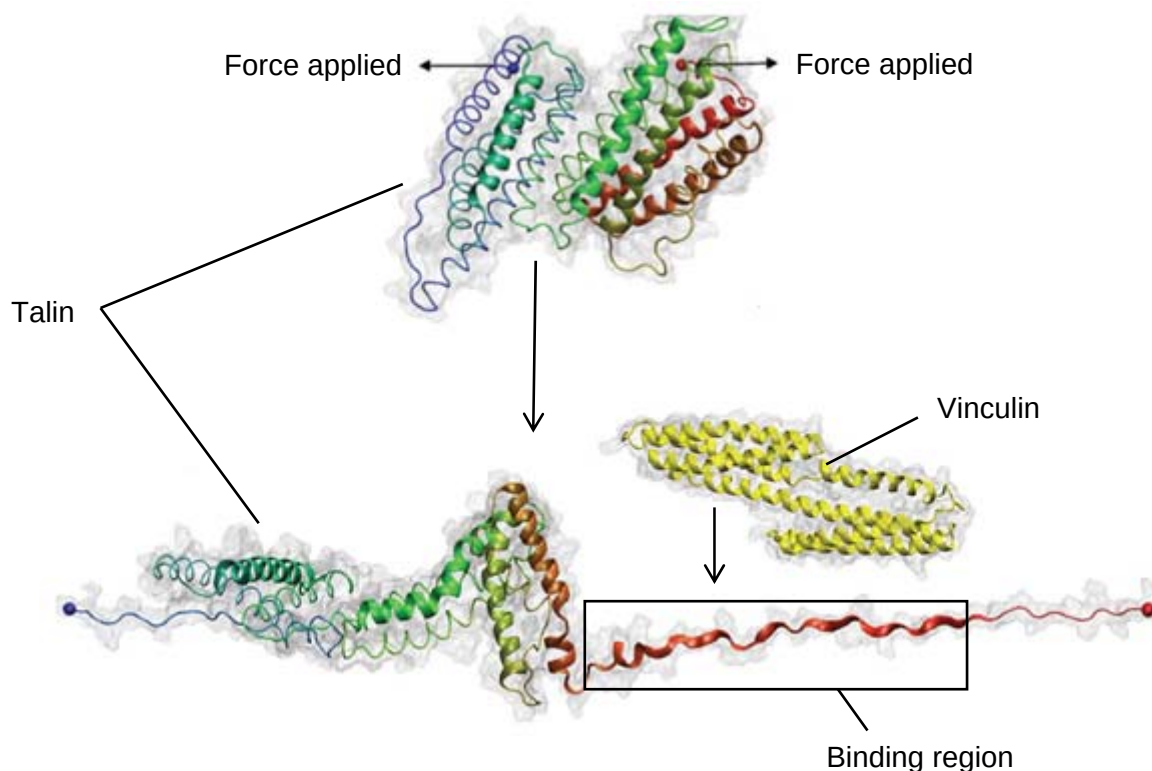


- 1 Which of the following are some visible structural features of the cell that enable it to function in secretion?

I	Presence of mitochondria to provide energy in the form of ATP for the process of secretion.
II	Presence of Golgi apparatus for the packaging and modification of proteins.
III	Presence of bound ribosomes for the synthesis of secretory proteins.
IV	Presence of numerous lysosomes for the digestion of macromolecules.
V	Presence of peroxisomes to aid in the process of transport of proteins to the cell surface membrane.

- A I and III only
 B II and IV only
 C I, II and III only
 D All of the above

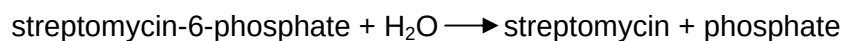
- 2 The micrograph is magnified 40,000X. Calculate the actual longest length of the nucleus.
- A $1.3 \times 10^{-5}\text{m}$
 B $3.1 \times 10^{-6}\text{m}$
 C $3.2 \times 10^{-7}\text{m}$
 D $3.1 \times 10^{-9}\text{m}$
- 3 The diagram shows how the 3D conformation of the protein talin can be altered to expose a binding region for another protein, vinculin, to bind.



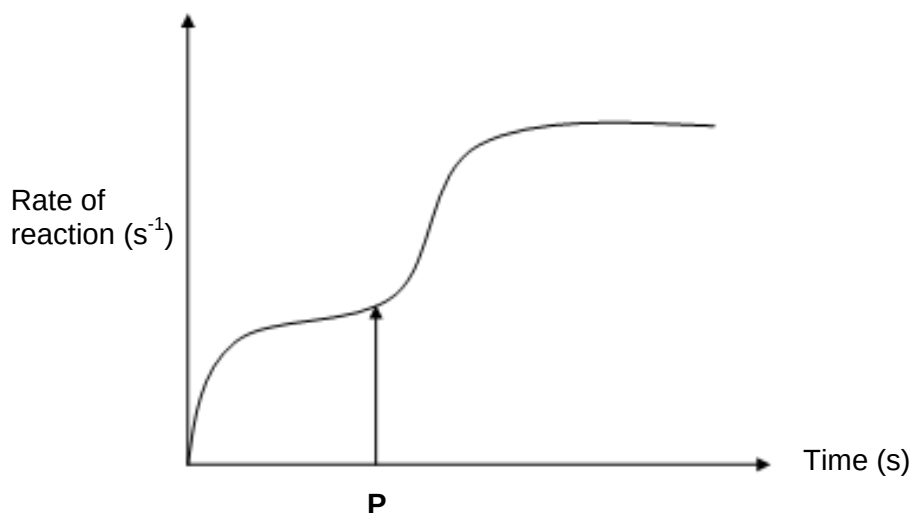
Which of the following changes must occur to the structure of talin to allow vinculin to bind to it when stretched?

	Primary structure	Secondary structure	Tertiary structure
A	No change	Ionic bonds broken	Hydrophobic interactions disrupted
B	No change	Hydrogen bonds broken	Ionic bonds broken
C	No change	Ionic bonds broken	Disulfide bonds disrupted
D	Peptide bonds broken	Hydrogen bonds broken	Ionic bonds broken

- 4 The enzyme streptomycin-6-phosphatase catalyses the final step in the production of the antibiotic streptomycin optimally at 30°C and pH 7.



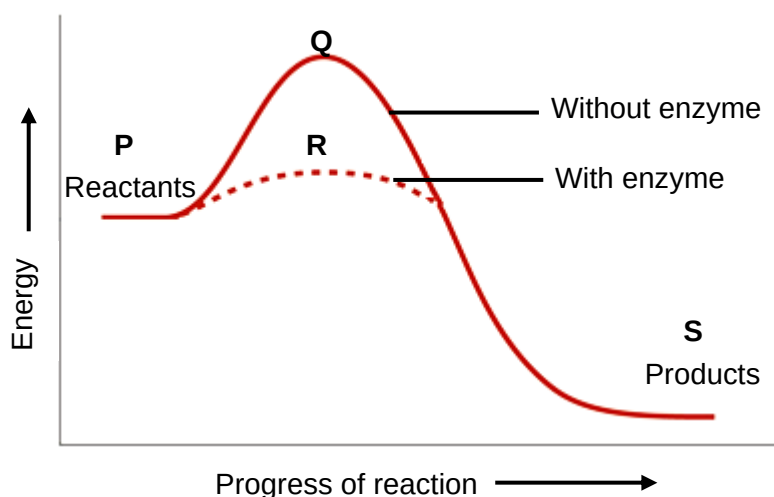
In one experiment, the time course of the reaction was followed by measuring the rate of reaction at regular time intervals, as shown in the graph.



Given that there was a change in one experimental condition at **P**, which of the following cannot be the change introduced?

- A Increasing temperature by 10°C from 15°C.
- B Doubling streptomycin-6-phosphate concentration.
- C Increasing pH from 7 to 10.
- D Doubling streptomycin-6-phosphatase concentration.

- 5 The effect of addition of an enzyme on the energy required for a reaction to proceed is shown in the graph.

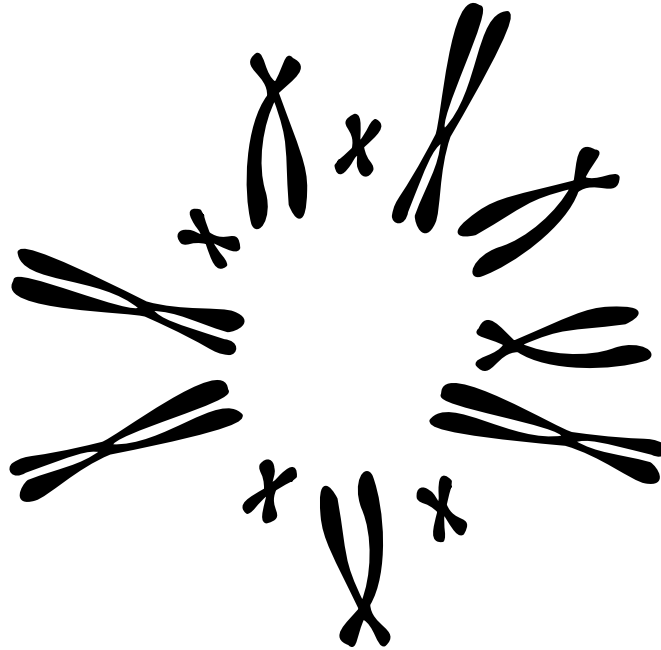


Which of the following statements accurately describe the events occurring at **P**, **Q**, **R** and **S**?

I	Addition of enzymes to a reaction would favour the formation of more products at S than the uncatalysed reaction.
II	Increasing the concentration of reactants at P would not result in a change in the energy values at Q and R .
III	Increasing the temperature of the reaction up to the optimum would not reduce the energy difference between Q and R .
IV	There would be a smaller number of molecules at R than at Q .

- A** I and II only
B II and III only
C III and IV only
D All of the above

- 6 The diagram shows the chromosomal behaviour in an animal cell during a particular stage of cell division.



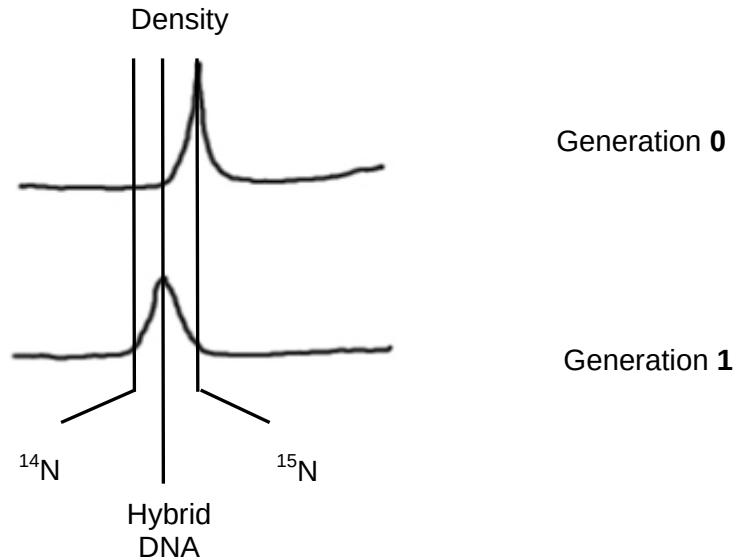
Which of the following correctly describes the chromosomes shown?

	Stage of cell division	Number of chromosomes in a haploid set	Total number of DNA strands
A	Metaphase	4	48
B	Prophase	2	24
C	Prophase	3	48
D	Metaphase	3	48

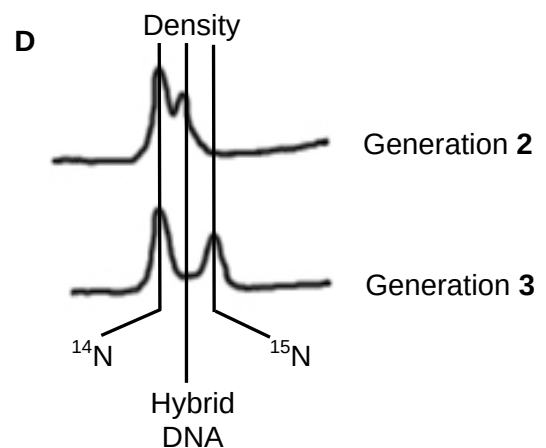
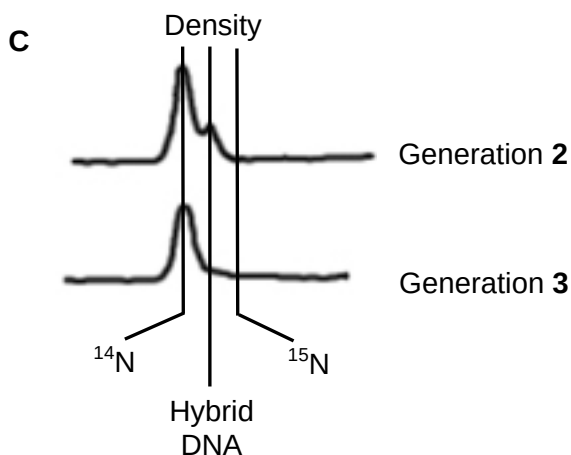
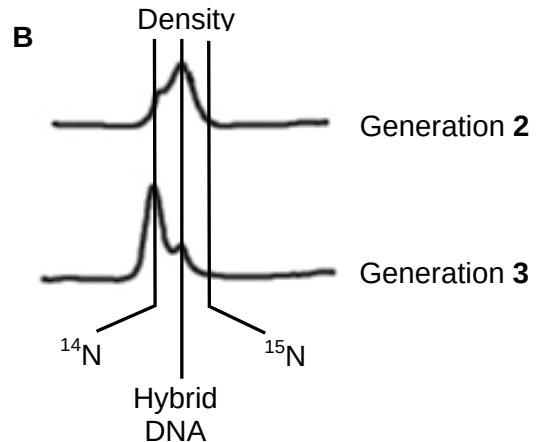
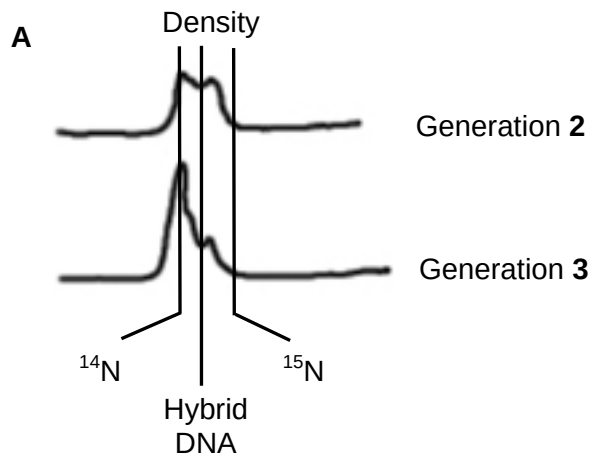
- 7 Which of the following is **not** true of adult stem cells during tissue repair?
- A The stem cells must have active telomerase.
 - B The different checkpoints in the cell cycle of the stem cells are activated.
 - C Mitosis of the stem cells is induced without any stimulus.
 - D The stem cells will stop dividing after the damaged cells are replaced.
- 8 A student designs an artificial DNA segment for insertion into the genome of a thermophilic bacterium. Which of the following DNA segments would be the most suitable for insertion?

A	CAACTAGGTT GTTGATCCAA	C	GAACTATCTT CTTGATAGAA
B	CAACTAGGTT GTTGATGCAT	D	GTTCTTGACA CAAGAAGAGT

- 9 The bacterium *Escherichia coli* was grown for many generations in ^{15}N and then subsequently transferred to ^{14}N to grow. Samples of *E. coli* were removed from the ^{14}N culture after each generation. The DNA from each sample was extracted and centrifuged. The diagram shows the results of the experiment for generations 0 and 1. The height of each peak represents the level of radioactivity.



Which of the following diagrams shows the correct densities and relative proportions of the DNA extracted from the *E. coli* at generations 2 and 3?



10 Which of the following statements describing tRNA are true?

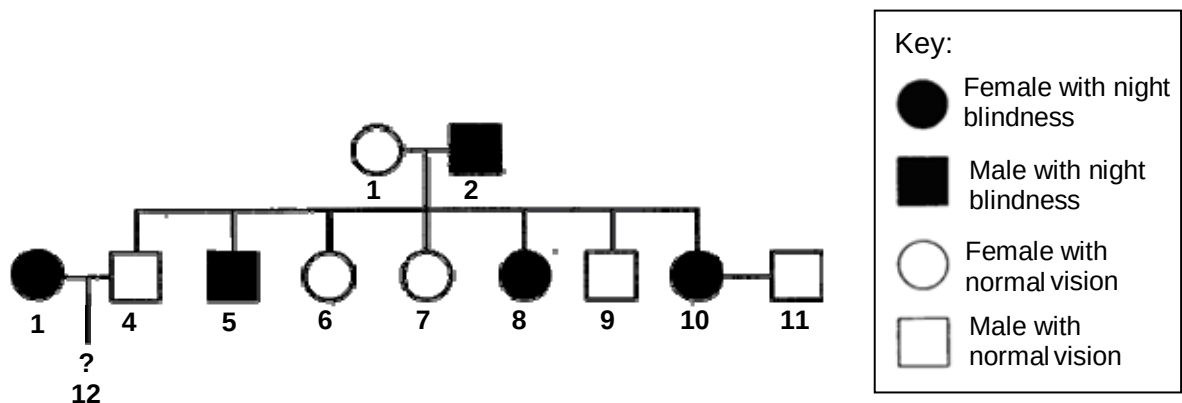
I	It is a polyribonucleotide chain that is folded into a precise 3D conformation stabilised by hydrogen bonds.
II	Its 3' end allows a specific amino acid molecule to bind to it via an ester linkage.
III	One of its loops has a specific triplet base sequence that is complementary to the mRNA sequence.
IV	It has a long poly(A) tail to confer stability in the cytoplasm.

- A I and II only
 B I, II and III only
 C I, III and IV only
 D All of the above

Questions 11 and 12 refer to the family tree.

Night blindness is a condition in which affected people have difficulty seeing in dim light. The allele **N** for night blindness is dominant to the allele **n** for normal vision.

The diagram shows part of a family tree illustrating the inheritance of night blindness.



11 Which of the following is a possible genotype for individual 12?

- A $X^N X^n$
 B $X^N Y$
 C nn
 D NN

- 12** What is the probability that the next two children born to individuals **10** and **11** will be an elder girl and a younger boy both with night blindness?
- A** 0.0625
 - B** 0.125
 - C** 0.25
 - D** 0.50

- 13** The type of plumage (feathers) in chickens results from the expression of an autosomal gene locus that carries two alleles:

Allele **H** (hen-feathering) is dominant in both female and male chickens.

Allele **h** (cock-feathering) is recessive, and the phenotype is expressed only in male chickens.

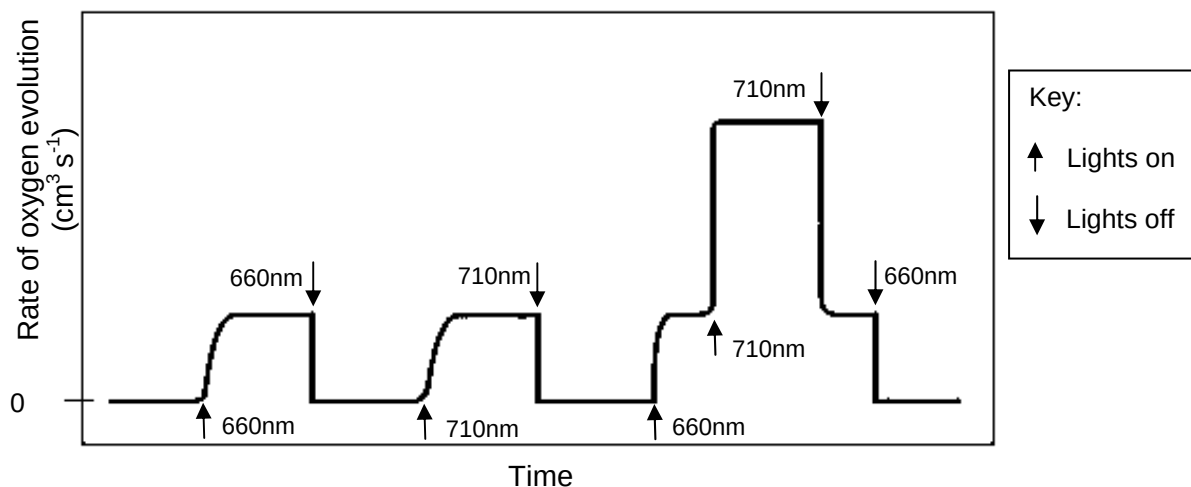
Which of the following would be the result of a mating between a cock-feathering male chicken, and a hen-feathering female chicken which is heterozygous at the gene locus?

- A** The probability of a male offspring being cock-feathering is 100% and that of a female offspring being hen-feathering is 100%.
 - B** The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 100%.
 - C** The probability of a male offspring being cock-feathering is 50% and that of a female offspring being hen-feathering is 50%.
 - D** The probability of a male offspring being hen-feathering is 100% and that of a female offspring being hen-feathering is 100%.
- 14** In humans, cataracts in the eyes and fragility of the bones are caused by dominant alleles found on autosomes that assort independently. Mr **X** who has cataracts but normal bones marries Miss **Y** who has fragile bones but normal eyes. Mr **X**'s father had normal eyes while Miss **Y**'s mother had normal bones.

What is the probability that the first child of Mr **X** and Miss **Y** will be free from both abnormalities?

- A** 0
- B** 0.25
- C** 0.5
- D** 1

- 15 Green algae were illuminated with two beams of light of wavelengths 660nm and 710nm, either singly or simultaneously as shown in the graph. The rate of oxygen evolution was subsequently measured.



Which of the following can be concluded from the graph?

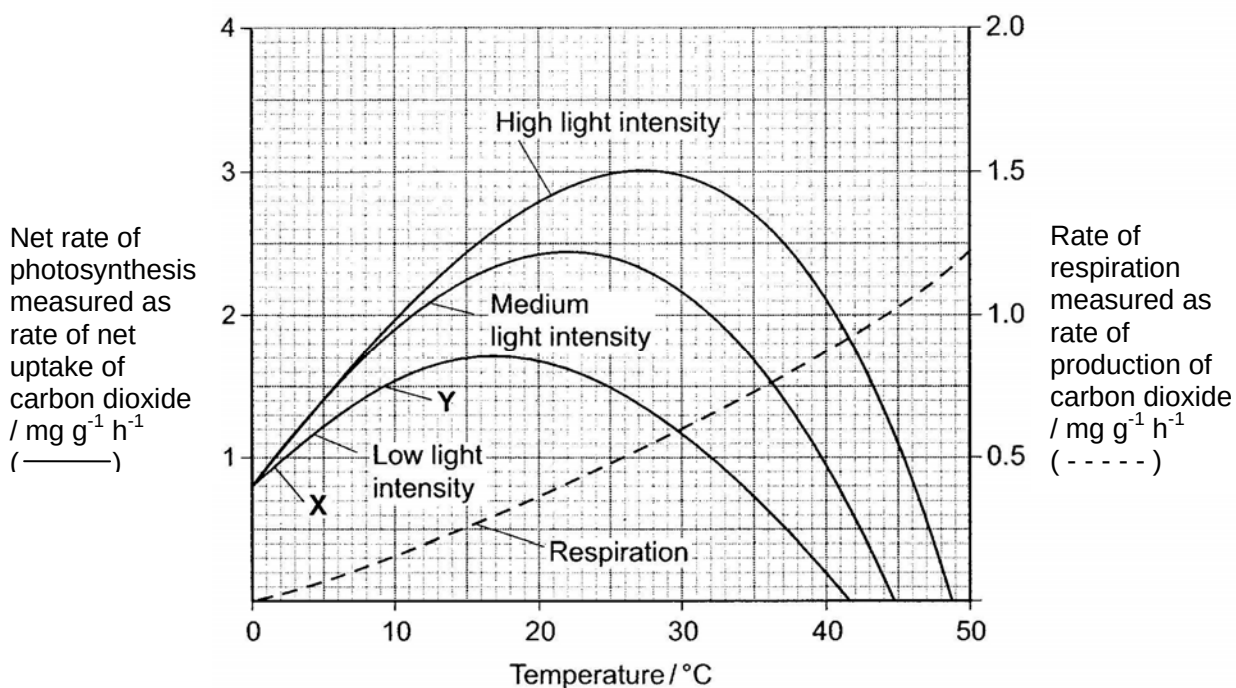
- A Photosynthesis is independent of both light wavelengths of 660nm and 710nm.
 B The rate of photolysis of water is greater at wavelength of 660nm than 710nm.
 C The photosystems of the green algae harvest light most efficiently at an average wavelength of 655nm.
 D Photosynthesis occurs maximally in the presence of both light wavelengths of 660nm and 710nm.
- 16 In a laboratory, a plant with variegated leaves (containing orange and green pigments) was supplied with radioactive carbon dioxide, $^{14}\text{CO}_2$. The plant was kept in the dark for 12 hours and then illuminated for the next 12 hours. A leaf from the plant was obtained and its level of radioactivity was measured both in the absence and presence of light. The results are shown in the table.

	Level of radioactivity in leaf (Arbitrary units)	
	Orange region of leaf	Green region of leaf
Absence of light	225	225
Presence of light	410	9271

Which of the following could be the most likely explanation for the level of radioactivity found in the orange region of the leaf in the presence of light?

- A Some photosynthesis occurs in the orange region but due to the absence of chlorophyll in that region, the rate of photosynthesis is low.
 B Photosynthesis occurs in the orange region but no storage of starch occurs.
 C Photosynthetic products diffuse into the orange region.
 D Radioactive $^{14}\text{CO}_2$ diffuses into the orange region and accumulates in that region.

- 17 Botanists studied the effect of light intensity and temperature on the rates of photosynthesis and respiration in the goldfish plant *Alloplectus nummularia*. The results are shown in the graph.

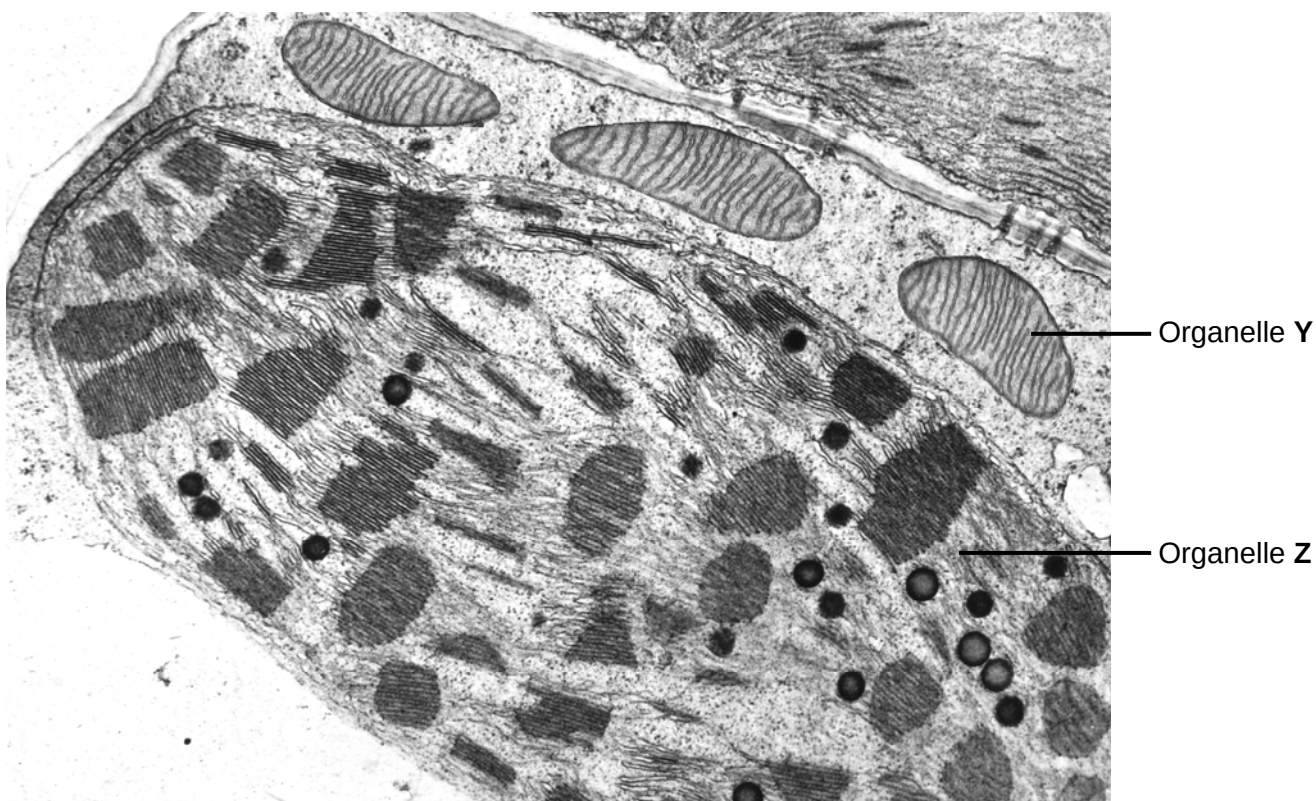


Which of the following statement(s) is / are **not** true?

I	Only light intensity limits the rate of photosynthesis between X and Y.
II	Unlike photosynthesis, the rate of respiration is dependent on temperature.
III	Under low light intensity, the compensation point of <i>A. nummularia</i> occurs at 30°C .
IV	Increase in temperature causes an increase in the rate of production of carbon dioxide by respiration.

- A IV only
- B I and II only
- C I, II and III only
- D All of the above

- 18 A new species of plant was recently discovered in the Amazon forest. The electron micrograph shows two organelles Y and Z in a leaf mesophyll cell of the plant.

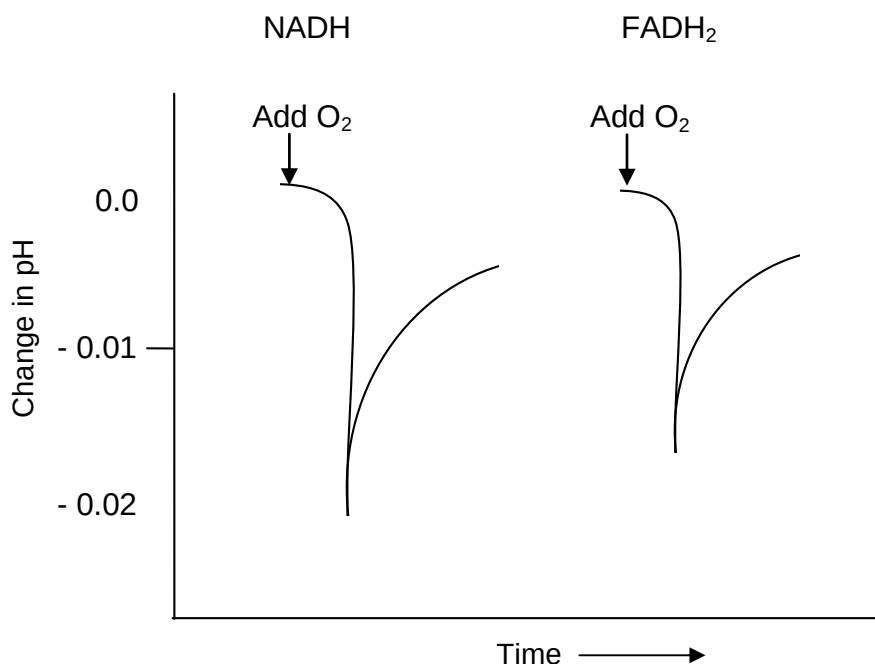


Which of the following statements are **not** true about organelles Y and Z?

I	Organelle Z utilises transporters to export ATP to organelle Y to drive cellular activities.
II	Oxygen released by organelle Z is used in organelle Y during Krebs cycle.
III	Phosphate ions in organelle Y is used for the production of ATP during Calvin cycle.
IV	NADPH molecules produced in organelle Z are used in organelle Y for the production of triose phosphate.

- A I and IV only
 B II and III only
 C II and IV only
 D All of the above

- 19 Isolated mitochondria were incubated with NADH in one experiment and an equal amount of FADH_2 in another experiment. The mitochondria were initially deprived of oxygen. The pH of the intermembrane space was then monitored as a known quantity of oxygen was added. The results are shown in the graph.

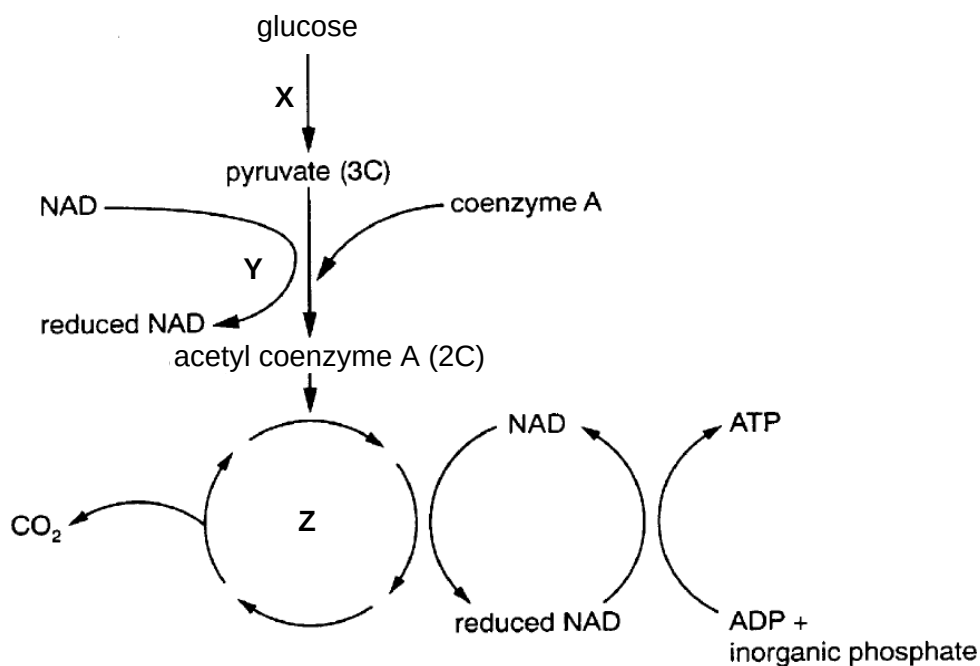


Which of the following can be concluded based on the results?

I	Upon the addition of oxygen, glycolysis and subsequently, link reaction, Krebs cycle and oxidative phosphorylation occurred.
II	Electron transfer was initiated by the addition of oxygen.
III	The pH drop was greater with NADH than with FADH_2 , which is consistent with the greater ATP yield that accompanies the oxidation of NADH.
IV	The rapid decline in pH indicates that protons were pumped into the intermembrane space when oxygen was available.

- A I only
 B II and IV only
 C II, III and IV only
 D All of the above

- 20 The flow chart shows a series of reactions occurring in an animal cell.



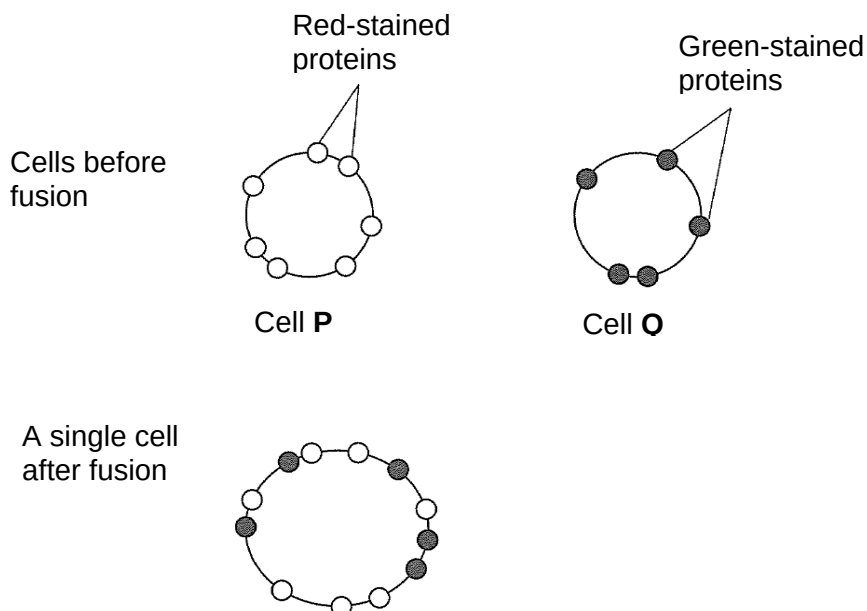
Which of the following statements correctly describes the flow chart?

- A Reaction X, which occurs in the cytosol, is an anabolic reaction.
- B Reaction Y involves the process of substrate-level phosphorylation, whereby pyruvate is first converted to a compound called acetyl coenzyme A.
- C Reaction Y occurs in the cytoplasm whereas reaction Z occurs in the mitochondria.
- D Reaction Z is a catabolic pathway which occurs twice for every glucose molecule to be completely oxidised.
- 21 The cell membrane of an African desert rat and that of the Antarctic penguin are compared to investigate their composition. Which of the following is the most probable makeup of the respective membranes?

	Amount of unsaturated fatty acids		Amount of cholesterol	
	Penguin	Desert rat	Penguin	Desert rat
A	Low	High	Low	High
B	High	Low	Low	Low
C	High	High	High	High
D	High	Low	High	High

- 22 In an investigation of the properties of the cell membrane, proteins in the membranes of Cell P and Cell Q were stained using different dyes. The proteins of one cell were stained green and the proteins of the other cell were stained red. The two cells were then fused to form a single cell.

The diagram shows the distribution of the proteins in the cell membranes before and after fusion.



Which of the following do the results illustrate?

A	Weak hydrophobic interactions between phospholipids allow fluidity in the cell membrane.
B	Fluidity of lipid bilayer allows the membranes to fuse.
C	Fluidity of lipid bilayer allows proteins to move within the cell membrane.
D	The cell membrane has a mosaic of various proteins embedded within it.

- 23 Which of the following statements is **not** true about transport across membranes?
- A** The steeper the concentration gradient, the greater the rate of diffusion.
 - B** Active transport is the facilitated movement of molecules against its concentration gradient, and requires energy in the form of ATP.
 - C** Facilitated diffusion is a form of passive transport that does not require energy.
 - D** In an animal cell, osmosis is the net movement of water molecules from a region of less negative solute potential to a region of more negative solute potential.

- 24** Which of the following conditions are commonly associated with evolution by natural selection?

I	Heritable variation
II	Competition for resources
III	Differences in fitness between individuals
IV	Major environmental changes

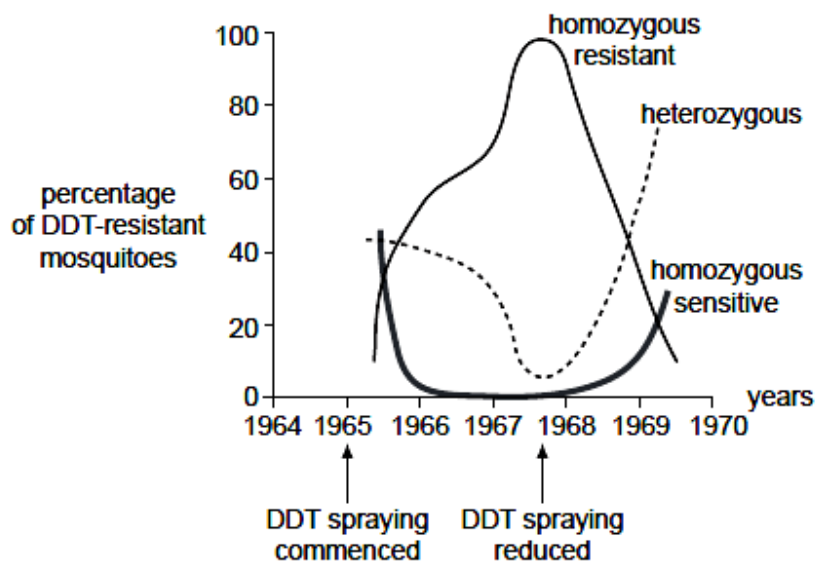
- A** II and IV only
B III and IV only
C I, II and III only
D I, III and IV only
- 25** Which of the following statements best describes the effect of genetic drift on the gene frequencies in a population?
- A** Genes enter a population by chance through immigration, thus changing gene frequencies.
B Chance alone can cause significant changes in gene frequencies of small populations.
C Mutations over time cause gene frequencies to change.
D Selection against one allele causes gene frequencies to change.

- 26** In the mid-1960s, DDT was widely used as an insecticide against mosquitoes. The sensitivity to insecticide in mosquitoes is determined by a single gene that has two alleles:

Allele 1 confers resistance to DDT.

Allele 2 confers sensitivity to DDT.

Genotypic frequencies were measured over several years in a population of mosquito larvae. The graph shows the results.



The results reveal that in the population of mosquito larvae,

- A** heterozygote advantage occurred when spraying levels declined.
- B** there were no alleles for sensitivity present in the population in 1967.
- C** the number of alleles for resistance was equal to the number for sensitivity in 1966.
- D** the homozygous resistant genotype was unable to produce offspring at low spraying levels.

- 27 The table shows the functional properties of three thermostable DNA polymerases (Pol) used in the Polymerase Chain Reaction (PCR).

Enzyme	Relative efficiency ^a	Error rate ^b	Processivity ^c	Extension rate ^d	3' to 5' exonuclease activity	5' to 3' exonuclease activity
<i>Taq</i> Pol	88	2×10^{-4}	55	75	No	Yes
<i>Tli</i> Pol	70	4×10^{-5}	7	67	Yes	No
<i>Pfu</i> Pol	60	7×10^{-7}	n.d.	n.d.	Yes	No

^a Percent conversion of template to product per cycle

^b frequency of errors per base pairs incorporated

^c Average number of nucleotides added before dissociation

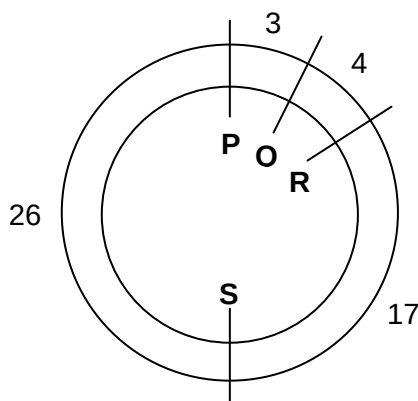
^d Average number of nucleotides added per second

n.d. not determined

Which of the following deductions is **not** true?

- A *Pfu* Pol is likely to have a lower extension rate than *Tli* Pol.
- B *Taq* Pol has proofreading capability.
- C The lower error rates of *Tli* Pol and *Pfu* Pol can be correlated with their 3' to 5' exonuclease activity.
- D *Taq* Pol has the highest relative efficiency as it has a comparatively high processivity and extension rate.

- 28 The diagram shows a plasmid with the positions of four restriction sites **P** to **S** indicated. The distances between them are shown in kb.



Copies of the plasmid were cut using two different restriction enzymes at a time, and the resulting fragments were separated by gel electrophoresis. The results are shown in the table.

Length of fragment (kb)	Restriction enzymes used		
	<i>BalI</i> and <i>EcoRI</i>	<i>PvuII</i> and <i>AvaI</i>	<i>PvuII</i> and <i>EcoRI</i>
40			
30			
20			
10			

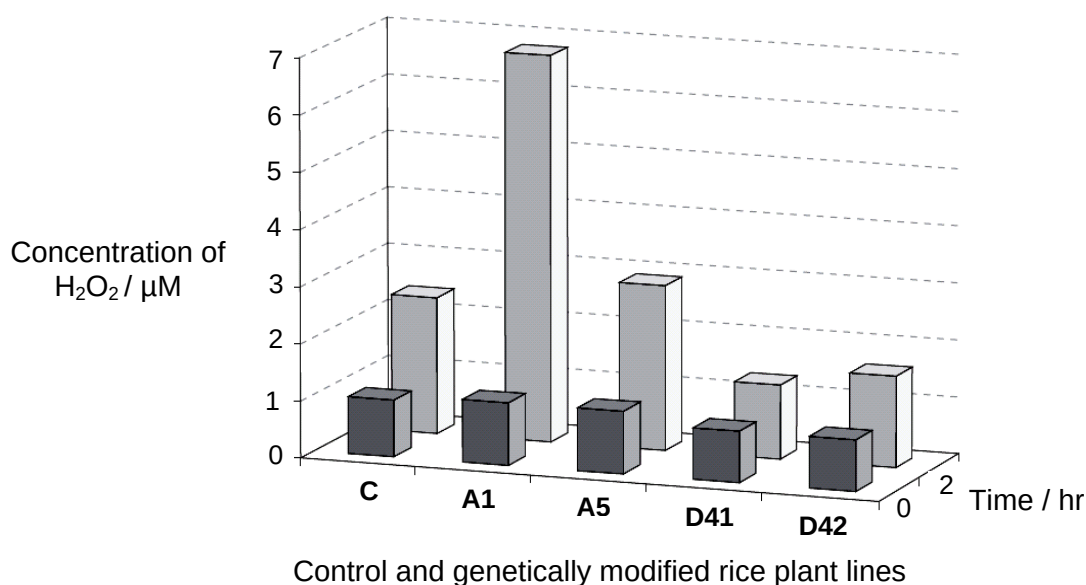
Which of the following correctly matches the four restriction enzymes to their respective restriction sites on the plasmid?

	P	Q	R	S
A	<i>AvaI</i>	<i>EcoRI</i>	<i>PvuII</i>	<i>BalI</i>
B	<i>BalI</i>	<i>AvaI</i>	<i>EcoRI</i>	<i>PvuII</i>
C	<i>PvuII</i>	<i>BalI</i>	<i>AvaI</i>	<i>EcoRI</i>
D	<i>EcoRI</i>	<i>PvuII</i>	<i>BalI</i>	<i>AvaI</i>

- 29** Which of the following statements about stem cells is true?
- A** Asymmetric cell division does not play a role in the formation of a cell lineage.
 - B** Only internal signals can trigger stem cell differentiation.
 - C** A progenitor cell has more limited developmental pathways compared to a multipotent stem cell.
 - D** Stem cell plasticity is the ability of a zygotic stem cell to develop into an adult stem cell.

- 30** Plants have developed defense mechanisms against pathogens such as bacteria, fungi, and viruses. Chemicals released by these pathogens can trigger defense responses in infected plant cells. One such response is the production of hydrogen peroxide (H_2O_2) which reacts with pathogen membranes and eventually kills both the pathogen and plant cells.

The *OSRac1* gene was isolated and introduced into a number of rice plant lines to study its role in disease resistance of plants to fungi. Experiments were carried out to determine if the *OSRac1* gene is part of the signalling pathway for H_2O_2 production. A control and four genetically modified rice plant lines **A1**, **A5**, **D41**, and **D42** were exposed to chemicals known to initiate H_2O_2 production. The results are shown in the graph.



Key:

C Control

A1 and A5 Rice plants with the *OSRac1* gene always turned on

D41 and D42 Rice plants with the *OSRac1* gene suppressed

Which of the following statements are true?

I	Production of H_2O_2 increased less in C plants than A1 and A5 plants.
II	Production of H_2O_2 increased less in D41 and D42 plants than A1 and A5 plants.
III	Transgenic plants showed more H_2O_2 production.
IV	Transgenic plants with gene suppressed showed less H_2O_2 production.

- A** I and II only
B II and III only
C I, II and IV only
D All of the above